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Gaussian Splatting-based Localization

MMML Final Project Report

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Abstract

We present a visual localization pipeline that estimates the 6DoF camera pose of a target RGB image using 3D Gaussian Splatting (3DGS) as the underlying scene representation. The method follows a two-stage design. First, it estimates an initial pose from a sparse Structure-from-Motion (SfM) point cloud using Hierarchical Localization (HLOC) components for global retrieval and local feature extraction, followed by Perspective-n-Point (PnP) with RANSAC. Second, it refines the pose through analysis-by-synthesis optimization by minimizing the photometric discrepancy between the query image and a view rendered from the explicit 3DGS map. To improve robustness, the pipeline incorporates candidate-specific PnP recovery and alternative initialization retries for high-loss cases. Our code is available at https://github.com/Vladick102/GS-based-Localization.

1 Introduction

Visual localization is the task of estimating a camera's position and orientation within a known environment. It is a fundamental problem in computer vision, robotics, augmented reality, and autonomous navigation.

Recent advances in neural scene representations, particularly Neural Radiance Fields (NeRF) [3], have demonstrated an ability to reconstruct photorealistic 3D scenes from images. However, NeRF-based methods are computationally expensive and often unsuitable for real-time applications. A newer paradigm, 3D Gaussian Splatting, offers a fast and efficient alternative.

This project focuses on visual localization using Gaussian Splatting-based scene representations. The goal is to analyze how explicit radiance representations built from anisotropic Gaussians can support accurate camera pose estimation.

1.1 Motivation

Compared to traditional Structure-from-Motion pipelines and NeRF-based approaches, Gaussian Splatting is computationally lighter while maintaining visual quality, making it a promising direction for practical deployment.

In the context of current conditions in Ukraine, robust vision-based localization is particularly relevant for autonomous systems operating in degraded or changing environments. Traditional discrete feature-matching algorithms often suffer from severe discontinuities and mathematical instability under such extreme appearance shifts.

Additionally, many existing works assume dense training views, static scenes, or controlled capture conditions. This project examines these limitations and assesses how they affect practical applicability.

Our work builds primarily on three research directions:

- 3D Gaussian Splatting as a scene representation [1]
- Visual localization methods based on Gaussian Splatting [2]
- Neural scene representation–based localization approaches [3, 4]

1.2 Research Questions

The project aims to investigate the following research questions:

- How suitable is 3D Gaussian Splatting as a scene representation for visual localization?
- What are the limitations of GS-based Localization?
- Can explicit Gaussian-based representations provide reliable localization in real-world scenarios?

1.3 Gaussian Splatting

Why. Kerbl et al. [1] address the need for a scene representation that combines high visual fidelity with real-time rendering speed. Previous neural representations, such as NeRF, provide high-quality reconstructions but require expensive volumetric sampling, which limits their practical use in time-critical applications.

How. 3D reconstruction from multi-view RGB images can be formulated as estimating a representation of a scene that allows the synthesis of images from novel viewpoints. In 3D Gaussian Splatting, the scene is modeled as a finite set of anisotropic Gaussian primitives. Each primitive is defined by parameters.

- Position (μ): A 3D vector representing the x, y, and z coordinates in world space; the mean of the Gaussian.
- Covariance matrix (Σ): A positive semi-definite matrix, representing orientation and size of the Gaussian primitive.
- Opacity (α): A single scalar value (0, 1).
- Color (c): View-dependent color is represented by Spherical Harmonics (SH) coefficients. The first degree of spherical harmonics is required to give splats a diffuse color, split into 3 channels (R, G, B). Additional view-dependent colors are supported through higher degrees of SH.

The spatial density of a Gaussian primitive is given by

$$G_i(\mathbf{x}) = \exp\big(-\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^\top \Sigma_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i)\big)\,,$$

which represents an ellipsoidal region of influence in 3D space.

Rendering is performed by projecting each Gaussian into the image plane. The color of a pixel is computed using alpha blending of splatted Gaussians ordered by depth.

The parameters of all Gaussians are optimized to minimize the difference between rendered images and ground-truth training views. Formally, given training images $\{I_k\}$ with known camera poses, the optimization problem can be written as

$$\min_{\{\boldsymbol{\mu}_i, \Sigma_i, \mathbf{c}_i, \alpha_i\}} \sum_k \mathcal{L}\big(R(\mathcal{G}, P_k),\, I_k\big),$$

where $\mathcal{G}$ denotes the set of Gaussian parameters, P_k is the camera projection for image k, R is the differentiable rendering operator, and $\mathcal{L}$ is typically a photometric loss (e.g., ℓ_2 loss).

What. The resulting method achieves real-time rendering of complex scenes while maintaining visual quality comparable to state-of-the-art neural rendering approaches. Instead of relying on multi-layer perceptrons to implicitly map spatial coordinates to radiance, 3DGS provides a continuous scalar field of densities that allows for exact, closed-form analytical gradients with respect to spatial parameters, avoiding the need for heavy numerical integration along camera rays.

So What. This representation makes neural scene models practical for applications beyond visualization, including localization, mapping, and robotics. Fast rendering and explicit geometry enable pose estimation and scene interaction in real time, motivating the study of Gaussian Splatting as a foundation for localization systems.

1.4 Localization

Visual localization is the problem of estimating the six-degree-of-freedom (6DoF) pose of a camera, consisting of rotation R and translation $\mathbf{t}$, with respect to a known scene representation. The pose can be written as a transformation matrix.

$$T = \begin{pmatrix} R & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{pmatrix}$$

Classical localization pipelines rely on detecting local features, matching them to a pre-built 3D map, and estimating pose via geometric algorithms. These methods degrade under appearance changes or sparse observations.

In representation-based localization, the scene model itself is used to evaluate candidate poses. For neural radiance fields, pose estimation is typically performed by minimizing photometric error between the observed image I and a rendered image from pose T:

$$\min_{T \in SE(3)} \mathcal{L}\big(\mathcal{R}(\text{scene}, T),\, I\big).$$

where $\mathcal{R}$ is a rendering operator.

Chen et al. [2] propose estimating camera pose directly from a Gaussian Splatting model using a single image. Such approaches demonstrate that explicit neural scene representations can serve not only for visualization but also as powerful geometric priors for localization.

2 Related Works

2.1 Structure-Based and Learning-Based Foundations

Traditional visual localization methods establish 2D-3D correspondences using point clouds constructed via Structure-from-Motion (SfM). These methods extract local features from the target image, match them against the 3D model's descriptors, and estimate the 6DoF camera pose using Perspective-n-Point (PnP) combined with RANSAC. They often fail in textureless or empty regions. Learning-based methods that encode scene information implicitly within neural networks, such as Absolute Pose Regression (APR) and Scene Coordinate Regression (SCR), aim to overcome this issue. However, these approaches often struggle with generalization across large-scale scenes and have low accuracy compared to direct geometric matching.

2.2 NeRF-Based Localization

The introduction of Neural Radiance Fields (NeRF) [3] and the large array of follow-up work have brought a new paradigm to novel view synthesis and subsequently to camera pose estimation. In particular, NeRF represents the scene as a continuous 3D volume and learns the radiance field properties, resulting in a more accurate and realistic map representation of static scenes with complex geometry and lighting. While NeRF-based camera pose estimation methods have certain advantages and can achieve competitive results, they also face limited applicability due to the major drawbacks, such as long model training and inference time. There are two main approaches for visual localization based on NeRFs:

2.2.1 Pose Refinement via Inverse Rendering

iNeRF [5] is the first method to directly optimize camera pose via inverse rendering, inspiring the development of subsequent methods. Given an initial camera pose guess, iNeRF iteratively refines the pose by minimizing the photometric error between the NeRF-rendered image and the target image. However, this analysis-by-synthesis approach is computationally expensive due to NeRF's backward ray-marching process. Additionally, it is highly prone to getting trapped in local minima, and strictly requires a relatively accurate initial pose prior to converge. To address the initialization limitation, methods like IFFNeRF [4] eliminate the need for a close initial pose guess by utilizing surface sampling and learned attention mechanisms.

2.2.2 Direct 2D-3D Feature Matching

To bypass the slow iterative rendering process of iNeRF, several methods extract 3D features directly from the NeRF model. CROSSFIRE [6] leverages dense local features from volumetric rendering to improve robustness. Similarly, NeRF-Loc [7] employs a conditional NeRF to generate continuous 3D descriptors for direct transformer-based matching with 2D image pixels.

2.3 Gaussian Splatting (3DGS) Based Localization

Recent advancements have introduced 3D Gaussian Splatting (3DGS) [1] as a fast, explicit alternative to NeRF. By representing the scene with anisotropic 3D Gaussians and utilizing a differentiable forward

rasterization pipeline, 3DGS achieves real-time rendering, making it an ideal candidate for visual localization. The methods initially proposed for NeRF-based representations have also migrated to Gaussian Splatting reconstructions:

2.3.1 Pose Refinement via Inverse Rendering

GSLoc [8] performs dense camera alignment by backpropagating pose gradients to minimize the photometric loss between the rendered and query images. To overcome the non-convexity of photometric losses that trap iterative solvers in local minima, GSLoc introduces a coarse-to-fine strategy using progressively decaying 2D Gaussian blur kernels on both images. Concurrently, LoGS [9] first estimates the initial pose via image retrieval and local feature matching with a PnP solver, followed by iterative refinement.

2.3.2 Direct 2D-3D Feature Matching

Operating without iterative rendering, STDLoc [10] extends 3DGS into "Feature Gaussians" and proposes a sparse-to-dense localization pipeline. It uses a matching-oriented sampling strategy to find 3D landmarks and a scene-specific detector to extract corresponding 2D features, solving the initial pose via PnP before doing dense feature alignment. Alternatively, 6DGS [2] eliminates the need for a pose prior by generating rays uniformly outward from Gaussian primitives. It utilizes an attention-based binding mechanism to solve the 6DoF pose using the rays via weighted Least Squares without iterative optimization.

3 Methodology

The localization approach presented in this work primarily builds upon the visual localization frameworks established by GSLoc [8] and LoGS [9]. The resulting system follows the same high-level design while incorporating several robustness-oriented modifications that improve stability in practice. In particular, retrieval and local feature extraction are performed through HLOC, pose initialization includes candidate-specific recovery in addition to aggregate PnP, and the refinement stage retains the best low-loss pose even if later optimization steps become unstable.

3.1 Method Overview

To provide a clear understanding of the proposed approach, we first outline the end-to-end localization pipeline. The system processes a target RGB query image to estimate its 6DoF camera pose through the following consecutive steps:

1. **Image Retrieval:** A global descriptor of the query image is computed and compared against a reference database to retrieve a subset of visually overlapping candidate frames. This stage uses the NetVLAD configuration provided by HLOC.

2. **2D-3D Matching:** Local keypoints are extracted from the query image and matched against the retrieved candidates to establish 2D-3D correspondences, leveraging the spatial coordinates from the underlying SfM point cloud. Local keypoints and descriptors are extracted with HLOC's SuperPoint configuration and then anchored to reference image observations that already have valid 3D SfM support.

3. **Pose Estimation via PnP:** An initial 6DoF camera pose (T_{init}) is calculated using a Perspective-n-Point (PnP) solver coupled with RANSAC to filter out geometric outliers.

4. **Refinement using Photometric Loss:** The initial pose is iteratively optimized by minimizing the photometric discrepancy between the target query image and the corresponding view rendered directly from the explicit 3DGS scene representation.

3.2 Initial Pose Estimation

To initialize the camera pose, we utilize the sparse point cloud generated via Structure-from-Motion (SfM) during the 3DGS training phase. Given a query image I_q, we first compute a global descriptor $\mathbf{v}_q \in \mathbb{R}^d$ and compare it against the descriptors of a reference database $\mathcal{D} = \{I_1, \ldots, I_N\}$, retrieving a subset of overlapping candidate images $\mathcal{S} \subset \mathcal{D}$.

Next, we extract a set of 2D keypoints $\mathcal{K}_q = \{\mathbf{p}_i \in \mathbb{R}^2\}$ and their localized descriptors from I_q. Because the candidate frames in $\mathcal{S}$ are already registered within the SfM framework, their 2D observations map to known 3D spatial coordinates $\mathbf{P}_j \in \mathbb{R}^3$. By matching the local descriptors of the query image to those of the candidate frames, we inherit these 3D coordinates and obtain the 2D-3D correspondences $\mathcal{M} = \{(\mathbf{p}_k, \mathbf{P}_k)\}_{k=1}^{K}$.

Correspondences from all retrieved candidates are first pooled and passed to a PnP solver with RANSAC. However, this aggregate set can become noisy when several nearby reference images provide partially inconsistent matches. To improve robustness, the system also evaluates PnP on strong reference candidates separately. If aggregate PnP fails, the best successful single-reference estimate is selected according to the inlier count. This extension proved important for stabilizing localization on our experimental scenes.

3.3 Iterative Pose Refinement

The initial pose T_{init} provides a strong starting point, but it still requires additional alignment. We refine the pose by minimizing the photometric discrepancy between the target query image I and the image rendered from the 3DGS map $\mathcal{G}$. Refinement uses differentiable rendering and automatic differentiation through the pose parameters.

Visual localization requires optimizing the camera pose matrix $T \in SE(3)$. The Special Euclidean group, $SE(3)$, represents the mathematical space of all 6DoF rigid-body transformations. It consists of a translation vector $\mathbf{t} \in \mathbb{R}^3$ and a rotation matrix $R \in SO(3)$. The Special Orthogonal group, $SO(3)$, defines valid 3D rotations, strictly requiring the 3×3 matrices to be orthogonal with a determinant of exactly 1.

Applying standard gradient descent directly to the transformation matrix T is unsuitable because additive updates will violate the geometric constraints of $SE(3)$ and $SO(3)$ (e.g., the rotation matrix will lose its orthogonality). To ensure a valid optimization, we parameterize the camera pose update using an unconstrained 6DoF vector $\boldsymbol{\xi} \in \mathbb{R}^6$. In this continuous representation, three components represent the spatial translation, and the other three represent the 3D rotation. $\boldsymbol{\xi}$ is unconstrained, therefore standard first-order optimizers like Adam can be applied to find the optimal pose update.

Once the optimizer computes the 6D update vector $\boldsymbol{\xi}$, it must be converted back into a valid 4×4 transformation matrix to update the camera pose. This conversion is performed using the matrix exponential map, denoted as $\exp(\boldsymbol{\xi})$. The exponential map wraps the unconstrained 6D vector into a strict rigid-body transformation matrix, intrinsically guaranteeing that the new rotation component remains perfectly orthogonal. The current pose is then updated by composing it with this new transformation matrix:

$$T_{\text{updated}} = \exp(\boldsymbol{\xi}) T_{\text{init}} \tag{1}$$

The optimization objective is to find the optimal update $\boldsymbol{\xi}^*$ that minimizes the photometric loss $\mathcal{L}$ between the ground truth image I and the rendered image:

$$\boldsymbol{\xi}^* = \arg\min_{\boldsymbol{\xi}} \mathcal{L}\Big(\mathcal{R}\big(\mathcal{G}, \exp(\boldsymbol{\xi}) T_{\text{init}}\big),\, I\Big) \tag{2}$$

where $\mathcal{R}$ is the 3DGS differentiable rendering operator. 3DGS provides a continuous scalar field of densities that allows exact analytical gradients to propagate backward from the rendered image pixels, through the projected 2D covariances, and directly into the extrinsic camera pose parameters.

Refinement is performed on a downscaled version of the query image whenever the original resolution is too large for stable and efficient optimization.

3.4 Optimization Strategies

A primary challenge in photometric pose optimization is the highly non-convex nature of the loss landscape, as the high-frequency image details often trap the optimizer in local minima. To mitigate this, two strategies can be applied:

- **Coarse-to-Fine Rendering:** Applying a progressively decaying 2D Gaussian blur kernel to both the rendered and target images during the initial optimization steps.
- **Pixel Masking:** To reduce the impact of noise and artifacts, construct binary masks to evaluate the photometric loss only on information-rich pixels, such as edges and keypoints, and regions with high opacity α.
- **Best-Pose Retention:** If the optimizer later diverges after already discovering a low-loss pose, the best stable pose is retained instead of discarding refinement altogether.
- **Alternative Initialization Retries:** If the refined result still has high photometric loss, the pipeline retries refinement from several alternative single-reference PnP initializations and keeps the best outcome.

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